QP Code:25900247

MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-UGP (HONOURS) Regular EXAMINATION October 2025

Third SEMESTER
(VAC) - MG3VACBTS200 – PUBLICHEALTH, HYGIENEANI
SANITATION : AN AWARENESS

(2024 ADMISSION ONWARDS)

Duration: 1.5 Hours **Maximum Marks: 50**

Part A

Fill in the blanks

Answer all questions

1. Critical thinking is the ability to analyze information in an objective manner. [K] / [CO3]
2. Vegetables should make up30-40% of daily diet. [K] / [CO2]
3. Dissolved oxygenis the amount of oxygen in water. [K] / [CO4]
4. Exercise can strengthen theintercostal muscles,improving lung functionand increasing oxygen intake. [K] / [CO1]
5. Healthy fatpresent in nuts,seeds,and avocados are an important part of a balanced diet. [K] / [CO2]
6. Unhealthy eating habits can contribute to lifestylediseases. [K] / [CO3]
7. Alcoholism is a chronic disease characterized by the inability to control alcohol consumption. [K] / [CO3]
8. Carbohydrates, protein andfatare classified as major nutrients. [K] / [CO2]
9. MPN is a method for determining drinking water quality. [K] / [CO4]
10. Water borne disease is a health effect of water pollution. [K] / [CO3]

[1x10 = 10]

Part B

Short Answer Questions

Answer any **10** questions

Each question carries **2** marks

11. Write the test for the detection of adulterant in turmeric powder. [K] / [CO3]

The test involves adding a few drops of hydrochloric acid to the powder, which will turn it pink if metanil yellow is present.

12. What is the procedure of secondary treatment of sewage? [K] / [CO4]

Trickling filter is used for the secondary treatment of sewage.

A trickling filter is a wastewater treatment process that uses a fixed bed of media, like rocks , for aerobic microorganisms to grow on as a biofilm. As wastewater is sprayed over the media, the biofilm consumes organic matter and nutrients, converting them into simpler forms. The treated water then flows out, and sloughed-off biofilm and solids are removed in a clarifier.

13. Write the mode of transmission of infant botulism. [K] / [CO4]

Infant botulism is transmitted when a baby ingests *Clostridium botulinum* spores, which can come from contaminated soil, dust, or honey. These spores germinate in the infant's immature digestive system, multiply, and release a toxin that affects the nervous system. Honey is a particularly well-known source of contamination for infants under one year old.

14. Write four benefits of physical activity. [K] / [CO1]

1.Improved physical health: Regular activity helps you maintain a healthy weight, lowers blood pressure, and reduces the risk of heart disease, stroke, and type 2 diabetes.

2.Enhanced mental well-being: Physical activity can improve mood, reduce feelings of stress, anxiety, and depression, and boost self-esteem. It can also lead to better focus and brain function.

3.Stronger bones and muscles: Exercise, especially weight-bearing activity, builds and maintains muscle strength, endurance, and bone density. This helps prevent conditions like osteoporosis and reduces the risk of falls.

4.Better sleep quality: Regular physical activity can help you4fall asleep faster and improve the quality of your sleep

15. List out the components of a healthy diet. [K] / [CO3]

components of a healthy diet

1.Carbohydrates: Choose complex and whole-grain carbohydrates like oats, brown rice, and whole wheat bread for sustained energy.

2.Proteins: Include lean meats, poultry, fish, eggs, beans, and nuts in your diet.

3.Fats: Healthyunsaturated fats from sources like vegetable oils, nuts, and seeds, while limiting saturated and trans fats.

4.Vitamins and Minerals: Food sources are fruits and vegetables.

5.Fiber: High-fibre foods include fruits, vegetables, whole grains, and legumes are important for healthy digestive system.

16. Write an analytical method to determine quality of drinking water. [K] / [CO4]

The Most Probable Number (MPN) test for coliforms involves serial dilution of a water sample and inoculating multiple tubes of a selective medium like lactose broth with measured volumes of each dilution. After incubation, a positive result is indicated by gas bubbles and acid production in the tubes, which is often detected by bubble in an inverted Durham tube and a color change from red to yellow in the medium. The number of positive tubes at each dilution is then used in conjunction with a standard statistical table to estimate the coliform count per 100 mL of the original sample.

17. Write two effects of exercise on skeletal system. [K] / [CO1]

Two effects of exercise on the skeletal system are increased bone density and improved joint health. Weight-bearing and resistance exercises stimulate bones to become denser, while strengthening the muscles around joints helps protect them from injury and improves stability.

18. What is malnutrition? [K] / [CO2]

Malnutrition refers to deficiencies or excesses in nutrient intake, imbalance of essential nutrients or impaired nutrient utilization.

19. Write two internal factors that affect health? [K] / [CO1]

1.Genetics: An individual's genetic makeup is inherited from their parents and plays a role in determining, overall healthiness. example, genetic predispositions can make some people more susceptible to conditions like diabetes.

2. Hormonal factors: Hormonal changes during different life stages can impact health.

20. Write the importance of sewage treatment. [K] / [CO4]

Sewage treatment is crucial for public health and environmental protection by removing harmful pathogens and pollutants before

wastewater is released. It protects human health by preventing waterborne diseases, safeguards ecosystems from pollution, and conserves freshwater resources through the reuse of treated water for irrigation and other non-potable purposes

21. Write two principles of accident prevention. [K] / [CO3]

Identifying the hazard, assessing the risk and controlling the risk are the principles of accident prevention.

22. Write two measures for preventing typhoid. [K] / [CO4]

1.Practicing good hygiene**.**

2.Consuming food that has been thoroughly cooked.

[2x10 = 20]

Part C

Short Essay Questions

Answer any **5** questions

Each question carries **4** marks

23. Describe the importance of fruits and vegetables in the d [K] / [CO2]

1.Rich in nutrients:

They are excellent sources of vitamins and minerals which are vital for many bodily functions.

2.Boosts immunity:

Vitamins and antioxidants found in fruits and vegetables help strengthen the immune system, making the body more resistant to infections and illnesses.

3.Helps digestion:

The high fibre content supports a healthy gut, helps prevent constipation, and contributes to overall digestive health.

4.Reduces disease risk:

Regular consumption is linked to a lower risk of serious health problems, including heart disease, stroke, type 2 diabetes, and certain types of cancer.

24. Discuss the benefits of probiotics for health. [K] / [CO3]

Probiotics are live microorganisms in foods intended to improve or restore microbiota in the gut. Probiotic food include yogurt, cheese, and clinical capsules containing specific probiotic strains.

Benefits of probiotics are

1.Gut health support

Taking probiotics naturally increases the amount of beneficial microorganisms in thegut.Probiotics also promote the production of antimicrobial substances, like short chain fatty acids and organic acids, and enhance the activity of certain immune cells, all of which help defend against harmful microbes.

Additionally, probiotics help improve the function and reduce the permeability of intestinal barrier, preventing harmful pathogens from entering the bloodstream.

Boost immunity

The balance of the gut microbiome plays an important role in both local intestinal immunity and systemic immunity.

Support mental health

The brain and the gut are connected through a two-way communication pathway called the gut-brain axis. This neurological connection makes it so that changes in the gut influence the brain activity.

25. What is potable water and why is it essential for human health? [K] / [CO4]

Potable water is water that is safe for human consumption, meaning it is free from harmful contaminants and suitable for drinking and cooking. Potable water is vital for health because it prevents waterborne diseases like cholera and dysentery, and is essential for numerous bodily functions including regulating body temperature, transporting nutrients, and flushing out toxins.

26. Explain the physical and mental dimensions of health. [K] / [CO1]

The physical and mental dimensions of health are intertwined, with **physical health** referring to the body's condition and its ability to function, and **mental health** encompassing the mind's well-being, including the ability to cope with stress and realize one's potential.

27. What are the harmful effects of junk food? [K] / [CO3]

1.Weight gain and obesity:

Junk food is high in calories, saturated fat, and sugar, which can lead to weight gain and obesity.

2.Cardiovascular problems:

High levels of sodium and unhealthy fats can lead to high blood pressure, high cholesterol, and an increased risk of

heart disease.

3.Type 2 diabetes:

The high sugar content in junk food can increase the risk of developing type 2 diabetes.

4.Cancer:

Regular consumption of junk food has been linked to an increased risk of certain cancers, including colorectal, stomachand respiratory cancers.

5.Liver disease:

Junk food can contribute to fatty liver disease.

7.Osteoporosis:
Some studies suggest a link between junk food and a higher risk of developing osteoporosis.

28. Discuss the importance of emotional expression and emotional self-care in one's life. [K] / [CO3]

Emotional expression and self-care are crucial for overall well-being, leading to better mental and physical health, increased resilience, and stronger relationships. Expressing emotions helps to process them and prevents negative impacts on health, while self-care practices build coping skills and emotional balance, making you better equipped to handle life's challenges.

Importance of emotional expression

1.Improves mental health: Articulating feelings helps in processing and regulating them, which can reduce stress and anxiety.

2.Prevents physical issues: Regularly expressing emotions can help end a fight-or-flight state, which is linked to various health problems.

Importance of emotional self-care

1.Enhances mental and physical health: Emotional self-care reduces the risk of mental health disorders, lowers stress, and can even improve physical health and recovery time.

2.Strengthens relationships: Emotionally balanced individuals tend to communicate more effectively, show empathy, and set healthier boundaries.

3.Increases productivity: Emotional well-being positively impacts the ability to concentrate, make decisions, and stay

motivated.

29. Explain the concept of the Eight limbs of Yoga. [K] / [CO3]

The eight limbs of yoga

1.Yama:

The first limb, focusing on ethical principles for relating to others.

2.Niyama:

The second limb, which involves personal observances and self-discipline.

3.Asana:

The physical postures, which are often the most recognized part of yoga in the West.

4.Pranayama:

The control and regulation of breath through various exercises.

5.Pratyahara:

The withdrawal of the senses from external stimuli to turn the focus inward.

6.Dharana:

The practice of focusing the mind on a single point, or concentration.

7.Dhyana:

The sustained and effortless state of meditation.

8.Samadhi:
The final limb, which is a state of profound absorption, oneness, and bliss, or enlightenment.